

# Grouping data

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عملية تصنيف البيانات الى مجموعات بناءا على مفتاح معين "Property معينة" Key.

Grouping refers to the operation of putting data into groups so that the elements in each group share a common attribute.

| Method Name | Description                                                                                                                   | C# Query Expression Syntax                                           |
|-------------|-------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| GroupBy     | Groups elements that share a common attribute. An <code>IGrouping&lt;TKey,TElement&gt;</code> object represents each group.   | <code>group ... by ... -or-<br/>group ... by ...<br/>into ...</code> |
| ToLookup    | Inserts elements into a <code>Lookup&lt;TKey,TElement&gt;</code> (a one-to-many dictionary) based on a key selector function. | Not applicable.                                                      |

Grouping is one of the most powerful capabilities of LINQ. The following examples show how to group data in various ways:

- By a single property.
- By the first letter of a string property.
- By a computed numeric range.
- By Boolean predicate or other expression.
- By a compound key.

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الفرق بين GroupBy و ToLookup

## 1. GroupBy

- Used deferred execution.
- Each iterate → group again (single process).

عشان هيا deferred execution ومش بتننفذ غير لما اعمل iterate عليها فهنا فكل مره انا هعمل process عليها هيبدا تنفيذا تاني عشان كد هتجنب استخدامها لو انا هعمل processes عليها في اكثر من مكان.

- Return `IEnumerable<IGrouping<TKey,TElement>>`.
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## 2. ToLookup

- Used immediate execution.
- Buffer the result in memory (multiple process).

هنا عشان immediate execution هي هتتنفذ مرة واحدة والداتا هتتخزن في الذاكرة مش محتاج يرجع تاني لمصدر البيانات ويبدأ يقسم تاني زي ال GroupBy فكد هستخدمها في حالات ال multiple process.

- Return `ILookup<TKey,TElement>`